Development Methodology

Hybrid AI Architecture & Performance Optimization Approach

Multi-Phase Development Strategy

Systematic approach combining web scraping, vector database optimization, and intelligent AI routing to create a production-ready customer support system with enterprise-level performance

WS Data Collection

1. **Selenium Web Scraping:** Icon-targeted content extraction from TradeUP FAQ website
2. **Structured Processing:** Custom QAJSONLoader for consistent data formatting
3. **Quality Validation:** Content cleaning and categorization across 16 topic areas

AI Intelligent Routing

1. **Hybrid Analysis:** Fast keyword detection combined with LLM intent classification
2. **Context Integration:** Conversation history and session data incorporation
3. **System Selection:** Optimal AI specialist routing based on query type analysis

SYS System Integration

1. **Modular Architecture:** Independent AI systems with shared routing layer
2. **API Integration:** Real-time market data and persistent memory systems
3. **Error Handling:** Graceful degradation and fallback response mechanisms

OPT Performance Optimization

1. **Multi-Level Caching:** Response cache, search cache, and session management
2. **Vector Optimization:** FAISS indexing with OpenAI embeddings for semantic search
3. **Parallel Processing:** ThreadPoolExecutor for concurrent operations

H

Hybrid Intelligence

Keywords + LLM analysis for optimal speed-accuracy balance

M

Multi-Modal AI

FAQ, market data, and memory systems in unified interface

R

Real-Time Integration

Live market data seamlessly combined with static content

S

Scalable Design

Production-ready architecture with enterprise performance

Methodology Results

Systematic development approach delivered a comprehensive AI system combining multiple specialized capabilities with intelligent routing and optimized performance

128+ Q&A Pairs

3 AI Systems

Smart Routing